

WI - 3.1/3.2 Reciprocal of Linear and Quadratic Functions

MHF4U

Jensen

1) Graph each of the following reciprocal functions. Start by graphing the function in the denominator. Show as much characteristic information about the graph as you can (e.g. intercepts, asymptotes with equations, other defining points, etc).

a) $f(x) = \frac{1}{x-1}$

HA: $y=0$

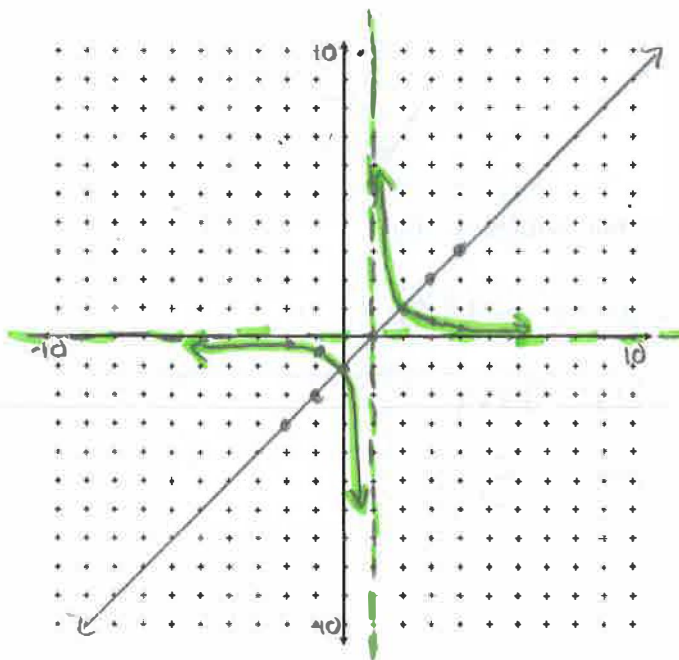
VA: $x=1$

$y=x-1$

x	y
-2	-3
-1	-2
0	-1
1	0
2	1
3	2
4	3

$f(x)$

x	$\frac{1}{y}$
-2	-0.33
-1	-0.5
0	-1
1	undefined
2	1
3	0.5
4	0.33



b) $g(x) = -\frac{2}{x+4}$

HA: $y=0$

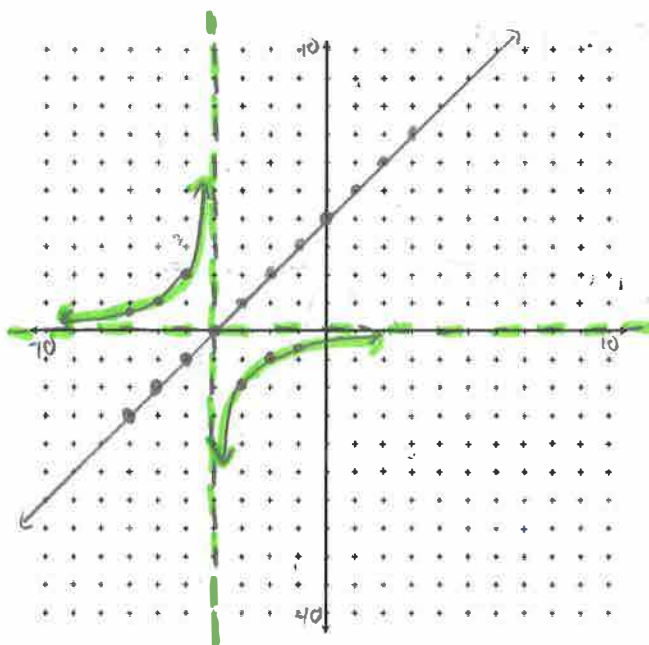
VA: $x=-4$

$y=x+4$

x	y
-7	-3
-6	-2
-5	-1
-4	0
-3	1
-2	2
-1	3

$g(x)$

x	$-\frac{2}{y}$
-7	0.67
-6	1
-5	2
-4	undefined
-3	-2
-2	-1
-1	-0.67



$$c) h(x) = \frac{1}{x^2-9} = \frac{1}{(x-3)(x+3)}$$

$$HA: y=0$$

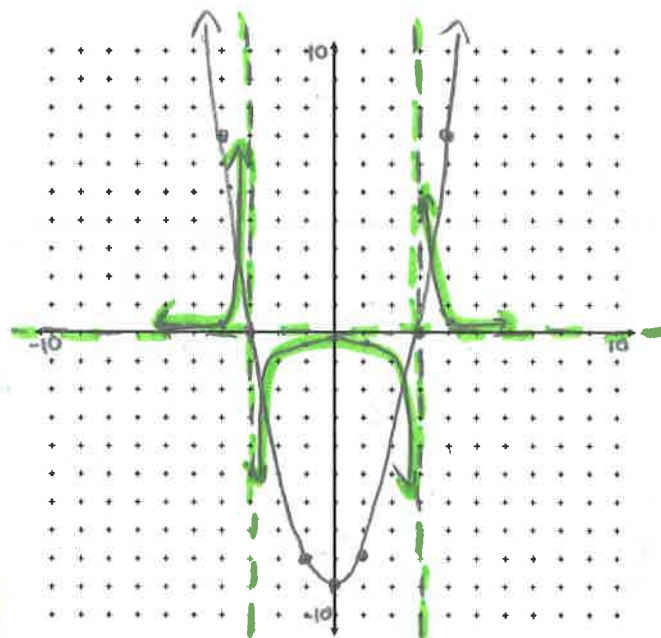
$$VA: x=3 \text{ and } x=-3$$

$$x^2-9 = (x-0)^2-9$$

x-vertex at (0, -9)

x	y
-4	7
-3	0
-1	-8
0	-9
1	-8
3	0
4	7

x	$\frac{1}{y}$
-4	$\frac{1}{7} = 0.14$
-3	undefined
-1	-0.125
0	-0.11
1	-0.125
3	undefined
4	0.14



$$d) j(x) = \frac{1}{x^2-2x-15} = \frac{1}{(x-5)(x+3)}$$

$$HA: y=0$$

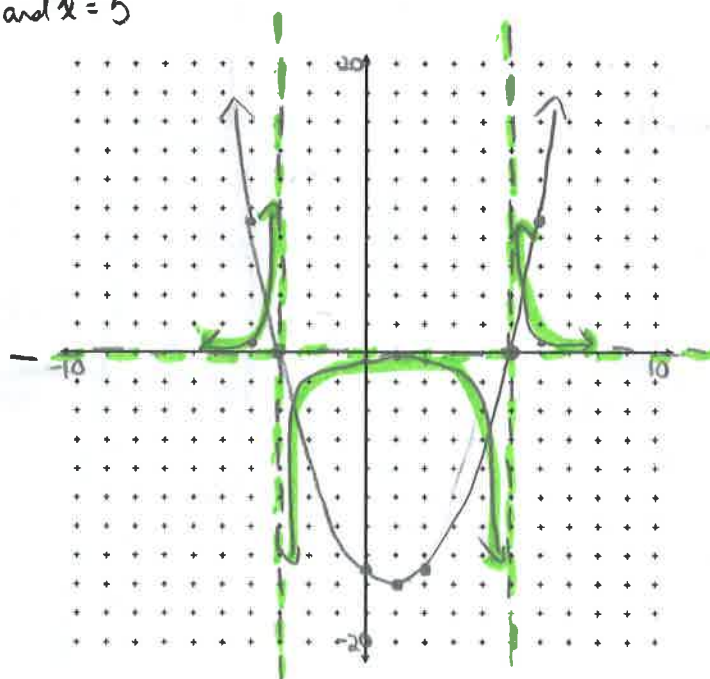
$$VA: x=-3 \text{ and } x=5$$

$$x^2-2x-15$$

x-vertex at $-\frac{b}{2a} = \frac{2}{2} = 1$

x	y
-4	9
-3	0
0	-15
1	-16
2	-15
5	0
6	9

x	$\frac{1}{y}$
-4	0.11
-3	undefined
0	-0.07
1	-0.06
2	-0.07
5	undefined
6	0.11



$$e) k(x) = \frac{1}{x^2+2}$$

$$HA: y = 0$$

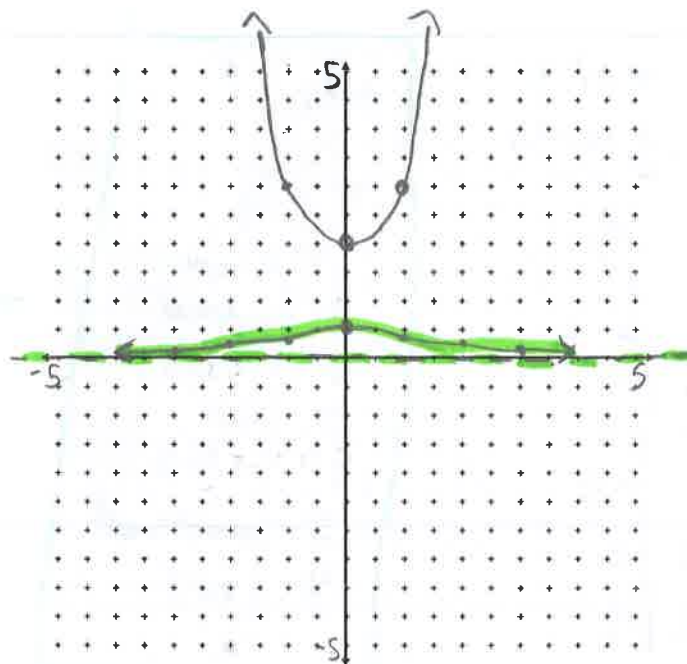
VA: NONE

$$x^2+2 = (x-0)^2+2$$

x-vertex at (0, 2)

x	y
-3	11
-2	6
-1	3
0	2
1	3
2	6
3	11

k(x)	
x	$\frac{1}{y}$
-3	0.1
-2	0.17
-1	0.33
0	0.5
1	0.33
2	0.17
3	0.1



$$f) m(x) = \frac{4}{x^2+x-6} = \frac{4}{(x+3)(x-2)}$$

$$HA: y = 0$$

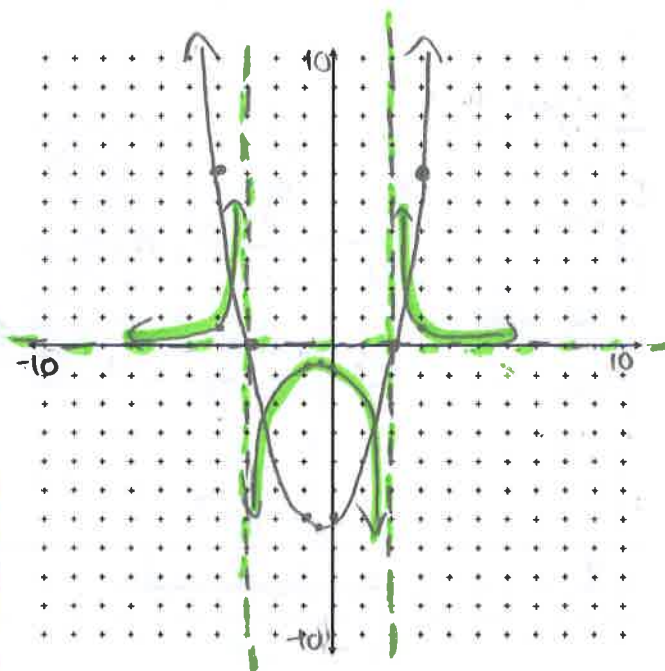
VA: $x = -3$ and $x = 2$

$$x^2+x-6$$

$$x\text{-vertex at } -\frac{b}{2a} = -\frac{1}{2(1)} = -0.5$$

x	y
-4	6
-3	0
-1	-6
-0.5	-6.25
0	-6
2	0
3	6

m(x)	
x	$\frac{4}{y}$
-4	0.67
-3	undefined
-1	-0.67
-0.5	-0.64
0	-0.67
2	undefined
3	0.67

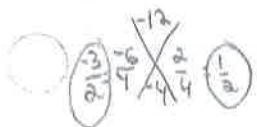


$$g) n(x) = -\frac{1}{4x^2 - 4x - 3}$$

$$= -\frac{1}{(2x-3)(2x+1)}$$

$$HA: y=0$$

$$VA: x=1.5 \text{ and } x=-0.5$$

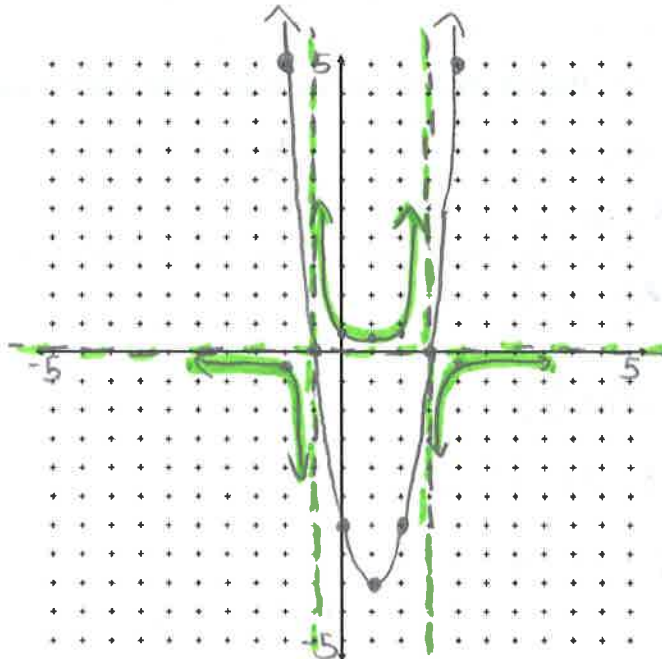


$$4x^2 - 4x - 3$$

$$x\text{-vertex at } -\frac{b}{2a} = \frac{4}{2(4)} = 0.5$$

x	y
-1	5
-0.5	0
0	-3
0.5	-4
1	-3
1.5	0
2	5

n(x)	
x	-1/y
-1	-0.2
-0.5	undefined
0	0.33
0.5	0.25
1	0.33
1.5	undefined
2	-0.2



$$p(x) = \frac{4}{2x^2 - 8x + 9}$$

$$b^2 - 4ac = (-8)^2 - 4(2)(9)$$

$$= -8$$

∴ No restrictions

$$HA: y=0$$

VA: None

$$2x^2 - 8x + 9$$

$$x\text{-vertex at } -\frac{b}{2a} = \frac{8}{4} = 2$$

x	y
-1	19
0	9
1	3
2	1
3	3
4	9
5	19

p(x)	
x	4/y
-1	0.21
0	0.44
1	1.33
2	4
3	1.33
4	0.44
5	0.21

